1. Power On
Power supplied to motherboard & CPU reset vector executed
2. BIOS / UEFI Initialization
POST (Power-On Self-Test) & hardware bus initialization
3. Boot Device Detection
Scan storage media (MBR for BIOS / ESP for UEFI)
4. Boot Loader (GRUB)
GRUB2 menu loads kernel (vmlinuz) & initramfs into RAM
5. Linux Kernel
Kernel boots, mounts temporary root FS & real root FS (/)
6. init / systemd (PID 1)
systemd starts as Process ID 1 to manage user-space
7. Services Start
Mount file systems, configure network & start SSH daemon
8. Login Prompt
Initialize virtual terminals & display login prompt (server01 login:)